

ARJUN CHANDRA

(+1) 435-625-9306

ac25@bu.edu

<https://arjunchandra2.github.io>

EDUCATION

Boston University

Boston, MA

B.S. in Physics & Computer Science

2021 - 2025

- GPA: 3.98/4.00
- Advisors: Boqing Gong, Venkatesh Saligrama, Irving Bigio
- Relevant Coursework:
 - Multimodal Machine Learning (Graduate)
Project: Targeted Caption Generation Improves Compositional Reasoning in VLMs
 - Bandit Algorithms (Graduate)
Project: Fair Algorithms for Multi-Agent Multi-Armed Bandits
 - Quantum Computing (Graduate)
Project: Lattice-Based Key Encapsulation Mechanisms in Post-Quantum Cryptography

WORK

Adobe | San Jose, CA

May 2024 – Aug 2024

EXPERIENCE

AI/ML Intern - Machine Learning Engineer

- Implemented and deployed a scalable LLM-based recommendation engine on roadmap for future release in Adobe Experience Platform AI Assistant
- Built and managed Intern Connect, a Slack app that pairs hundreds of Adobe interns around the world for weekly virtual coffee chats

RESEARCH

Boston University Data Science & Machine Learning

June 2025 – Present

EXPERIENCE

Research Fellow | Advisor: Venkatesh Saligrama

- Curating synthetic data and developing automatic evaluation methods to probe failure modes in speech to speech models

Boston University Image & Video Computing

Jan 2025 – Present

Research Assistant | Advisor: Boqing Gong

- Investigating developmentally plausible learning mechanisms for intelligent and data-efficient pre-training of multimodal foundation models
- Published second-author paper in ICCV 2025

Adobe Research

Aug 2024 – May 2025

Student Researcher | Advisor: Yunyao Li

- Integrated multi-source knowledge retrieval into text-to-video pipelines to achieve improved factual accuracy and interpretability
- Published first-author paper in AAAI 2025 and demonstration proposal in CVPR 2025

Boston University Biomedical Optics Lab

Aug 2023 – May 2025

Research Assistant | Advisor: Irving Bigio

- Implemented 3D object detection methods to identify myelin degeneration in microscopic brain imaging of late-stage Alzheimer's and CTE patients
- Published in Alzheimer's & Dementia Journal and first-author manuscript submitted to Neurophotonics

PUBLICATIONS	1. Daniel Lee*, Arjun Chandra*, Yang Zhou, Yunyao Li, Simone Conia. <i>Grounding Pixels in Facts: Distilled Knowledge Retrieval for Factual Text-to-Video Generation</i>. CVPR 2025 Demonstration Program. 2. Shengao Wang, Arjun Chandra, Aoming Liu, Boqing Gong, Venkatesh Saligrama. <i>BabyVLM: Data-Efficient Pretraining of VLMs Inspired by Infant Learning</i>. ICCV 2025. 3. Daniel Lee*, Arjun Chandra*, Yang Zhou, Yunyao Li, Simone Conia. <i>Rewind and Render: Towards Factually Accurate Text-to-Video Generation with Distilled Knowledge Retrieval</i>. AAAI 2025 Demonstration Program. 4. Anna Novoseltseva, Gulce Kureli, Shuaibin Chang, Jiarui Yang, Precious D. Antinew, Arjun Chandra, Samer Berghol, Elizabeth Spurlock, Ann C. McKee, Bertrand R. Huber, Hui Wang, David A. Boas, Irving J. Bigio. <i>Imaging myelin degradation in ex vivo pre-frontal cortex tissue blocks in Alzheimer’s disease and chronic traumatic encephalopathy</i>. Alzheimer’s & Dementia. 5. Anna Novoseltseva*, Arjun Chandra*, Alexander J. Gray, Shuying Li, Mikayla Bradsby, Irving J. Bigio. <i>Accelerating Myelin Defect Detection in Neurodegenerative Disorders: A Human-in-the-Loop Deep Learning Approach with Birefringence Microscopy</i> . Under Review.
TEACHING EXPERIENCE	• Teaching Assistant, CS 460 (Database Systems)Spring 2025 • Course Assistant, CS 460 (Database Systems)Fall 2024 • Learning Assistant, PY212 (General Physics 2)Fall 2023 • Learning Assistant, PY252 (Electricity & Magnetism)Spring 2023 • Peer Tutor, Educational Resource Center2022 - 2024
AWARDS & HONORS	• Faculty Award for Interdisciplinary Majors, Boston UniversityMay 2025 Awarded to one graduating senior in the interdisciplinary physics major for outstanding academic performance and significant contributions to interdisciplinary research. • UROP Student Research Award, Boston UniversityJanuary 2025 Undergraduate research funding awarded for three consecutive semesters. • SAEF Award, Boston UniversityDecember 2024 1 of 17 students in the College of Arts & Sciences awarded funding for conference travel and research expenses. • Third Place, MIT iQuHackFebruary 2024 Earned third place in MIT’s annual international quantum computing hackathon.
LEADERSHIP & OUTREACH	Boston University PRISMAug 2024 – May 2025 <i>Peer Mentor</i> • Mentored incoming class of physics majors in course planning, research, and other campus opportunities Boston University Children’s CenterJan 2024 - Aug 2024 <i>Classroom Assistant</i> • Assisted teachers with daily routines and educational activities for preschool children Boston University Music EngagementSept 2023 – May 2024 <i>Treasurer</i> • Coordinated with Allocations Board to manage semester budget and submit budget requests